



FLEET TEAM DIGEST¹

737MAX-FTD-46-19002²

Issue Title : Network File Server 2 (NFS-2) Hardware Related Issues³

Airplane Model	ATA	Minor Model(s)	
737MAX	4613-01	-7,-8,-8200,-9,-BBJ,	
Other Model(s)	ECCN		
	9E991		
Originated Date	Last Revised Date	Created On	
05/23/2019	04/10/2021	04/05/2021	
Estimated Completion Date	Next Update Date		

Revision Description⁵

[Revised Status Section, Final Action, and the Milestone Section with connector solder joint failures dates.](#)⁶

Applicability⁷

737 MAX airplanes with the NFS serial numbers (S/Ns) mentioned in the Status section below.⁸

Description⁹

737MAX operators have experienced Network File Server (NFS) issues. Currently, there are two known hardware related issues with the NFS: Solid State Drive (SSD) and Network Extension Device (NED) Subsystem Processor (NSP) connector failures.¹⁰

[SSD Failure Symptoms:](#)¹¹

- Fault Light on the front of the NFS is solid ON for greater than 10 minutes following power-up.¹²

[SSD Failure Background:](#)¹³

- Delkin (SSD supplier) failure analysis did not identify root cause.
- Delkin attempted to resolve the issue with a firmware update, which had no positive effect.
- The leading hypothesis is that the SSDs are susceptible to power interruptions while they are being written to.

[Connector Solder Joint Failure Symptoms:](#)¹⁵

- The typical failure is a degraded performance of the A429 ACARS Output (output #4). However as the failure mode progresses, this can become a persistent failure of this output port. In extreme cases, other A429 and Program Pins may be affected.¹⁶

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- No Present Leg Fault (PLF) and Real-Time Event (RTE) reports downlinked via ARINC Communications Addressing and Reporting System (ACARS) datalink.
- No CFM LEAP EEC reports downlinked to ground via ACARS datalink.
- Excessive instances (10 or more over the last 10 flight legs) of Maintenance Message 23-23010 (CMU HAS NO INPUT FROM NFS-L ON CABIN TERM #1).
- Not affected: PLF and Fault History (FH) reports are downlinked to Loadable Software Airplane Part Library (LSAPL) via Wireless Network or Gatelink
- Not affected: Able to access to the PLF, FH, and Inbound Flight Deck Effects from the Multi-function Display System (MDS) or Maintenance Laptop

Connector Solder Joint Failure Background: ³

- Service Related Problem 737MAX-SRP-46-0006 was opened to investigate the NSP Connector issue ⁴
- The preliminary investigation indicates the internal Circuit Card Assembly (CCA) connectors are being exposed to forces greater than design.
- Root Cause of the solder joint failures has been determined.

Status ⁵

SSD Status: ⁶

- Teledyne Controls selected an alternate supplier for the Solid State Drive (SSD): ATP Electronics.
- ATP Electronics SSD are newer technology with improved firmware and architecture.
- ATP Electronics SSD provide a power protection feature to ensure a graceful shutdown.
- The ATP Electronics SSD built in power protection feature does result with a requirement to fully power cycle the NFS during manual resets. If using the Circuit Breaker, ensure that it is pulled for a minimum of three seconds before reapplying power.
- Teledyne Controls replacing Delkin with ATP Electronics SSD on returned units with a failed Delkin SSD.
- Production units (beginning with S/N 47B01090) are being delivered with ATP Electronics SSD. Refer to the Service Bulletin for applicable serial numbers
- Field returns with replacement SSD are being tracked by Mod Dot and Serial Number.
- Teledyne Controls Service Bulletin (SB) SB2247200-46-7 was released for NFS P/N 2247200-01, and lists the NFS serial number effectivity.
- Teledyne will the following service bulletins by the end of 1Q 2021.
- [Teledyne Controls Service Bulletin \(SB\) SB2247200-46-12 is released for NFS P/N 2247200-01](#)
- [Teledyne Controls Service Bulletin \(SB\) SB2247200-46-15 is released for NFS P/N 2247200-03](#)

Connector Solder Joint Status: ⁸

- Maintenance Message 23-23010 (CMU HAS NO INPUT FROM NFS-L ON CABIN TERM #1) is effective ⁹ for predictive alerting of the connector issue.
- Teledyne performed a failure analysis and modified the design to mechanically disconnect the Network Extension Device (NED) Subsystem Processor (NSP) from the Server Subsystem Processor (SSP) Module Assembly.
- Root Cause has been established. NSP Connector: The mechanical tolerance stack-up in the NFS and the dissimilar thermal expansion properties of the epoxy used on the connectors resulted in mechanical stress causing fractures in the solder terminations between the connector and the printed wiring board.
- Teledyne will the following service bulletins by the end of 1Q 2021.

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- Teledyne SB SB2247200-46-13 supersedes SB2247200-46-9 and is released for NFS PN 2247200-01.²
- Teledyne SB SB2247200-46-14 is released released NFS PN and 2247200-03.

Interim Action³

If NFS exhibits symptoms described in the discussion section, remove the NFS per AMM Task 46-13-01-000-801 (Network File Server Removal) and Replace the NFS per AMM Task 46-13-01-400-801 (Network File Server Installation).⁴

Final Action⁵

Boeing recommends airline operators send failed NFS units back to Teledyne Controls for repair.⁶

Root Cause has been established. on the NSP connector a mechanical tolerance stack-up in the NFS and the dissimilar thermal expansion properties of the epoxy used on the connectors resulted in mechanical stress causing fractures in the solder terminations between the connector and the printed wiring board.⁷

Implement Teledyne SB SB2247200-46-13 for NFS PN 2247200-01, and Teledyne SB SB2247200-46-14 for NFS PN 2247200-03.⁸

Close this FTD⁹

Milestones¹⁰

Note: Target dates are based upon currently available status and have risk of future revision.¹¹

NFS2 SSD Replacement¹²

Root Cause Established: March 2019

Solution Selected: March 2019

Change Committed: March 2019

Service Bulletin Available: Teledyne Controls Service Bulletin (SB) SB2247200-46-12 is released for NFS P/N 2247200-01

Teledyne Controls Service Bulletin (SB) SB2247200-46-15 is released for NFS P/N 2247200-03

Parts Available: 18-March-2021

Boeing Service Letter: 737-SL-46-039-A (released)

Connector Solder Joint Issue¹³

Root Cause Established: October 2020

Solution Selected: December 2020

Change Committed: 1Q 2021

Service Bulletin Available: March 19, 2021.

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Teledyne SB SB2247200-46-13 will be released for NFS PN 2247200-01.²
Teledyne SB SB2247200-46-14 will be released for NFS PN 2247200-02, and 2247200-03.
[Parts Available: 18-March-2021](#)
[Boeing Service Letter: 737-SL-46-043 \(released\)](#)

References³

Reference Type	Reference Number	Revision Number	Reference Date	Actions
Service Bulletin (SB)	Teledyne Controls SB2247200-13, PN 2247200-01			
Service Related Problem (SRP)	737MAX-SRP-46-0006		12/01/2017	
Fleet Team Digest Article (FTD)	737MAX-FTD-46-19001			
Service Bulletin (SB)	Teledyne Controls SB2247200-46-7			
Service Bulletin (SB)	Teledyne Controls SB2247200-46-9			
Service Bulletin (SB)	Teledyne Controls SB2247200-12, PN 2247200-01			
Service Bulletin (SB)	Teledyne Controls SB2247200-15, PN 2247200-02, 03			
Service Bulletin (SB)	Teledyne Controls SB2247200-14, PN 2247200-02, 03			

Parts List⁴

Parts Name	Boeing Part Number	Supplier Name	Supplier Number	Actions ⁵
Network File Server 2		Teledyne	2247200-01	
Network File Server 2		Teledyne	2247200-03	

Related Categories⁶

- Component Issues - Issues that occur during component maintenance or have an impact on component reliability⁷
- Dispatch Reliability Issues - Issues that cause revenue flight delays and cancellations
- Fleet Team IdeaXchange - Topic of discussion on the Fleet Team IdeaXchange bulletin board

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